

Project by Avrillia Carolyn Sidabutar

Data Cleaning

Sensitivity Analysis

Power BI

CAFE SALES: FROM 10,000 MESSY ROWS TO A DASHBOARD YOU CAN TRUST

A YEAR OF CAFE TRANSACTIONS WHERE 69% OF ROWS HAD AT LEAST ONE MISSING OR INVALID FIELD. I PROFILED THE PROBLEMS, FIXED WHAT COULD BE FIXED, AND TESTED WHETHER MY ASSUMPTIONS CHANGED THE CONCLUSIONS.

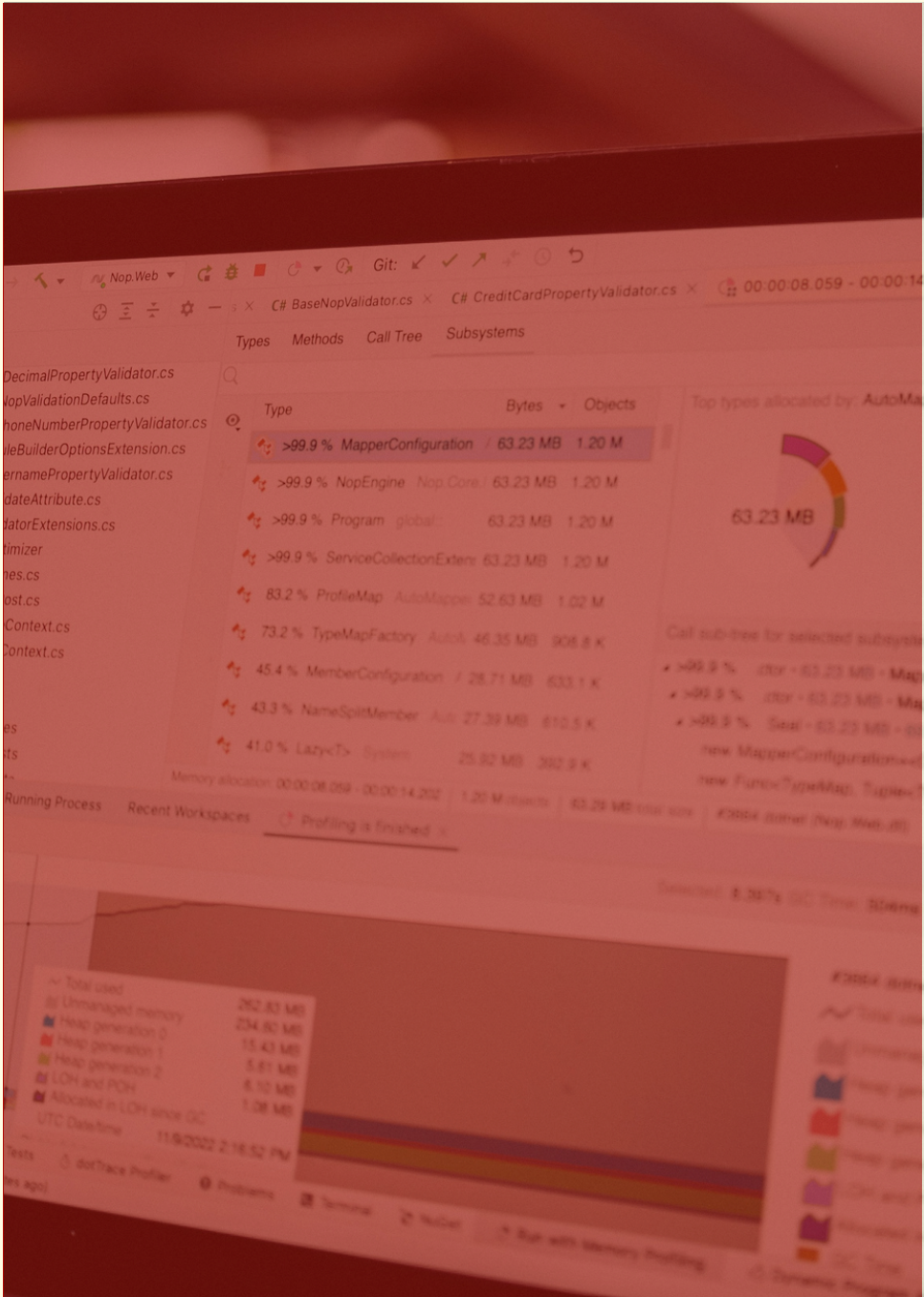
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DATASET INTRODUCTION

Column name	Description
transaction_id	a unique transaction identifier
item	menu item sold
quantity	number of units purchased in the transaction
price_per_unit	retail price for a single unit of the item
total_spent	gross monetary value spent on the transaction
payment_method	payment channel used
location	dining fulfillment location
transaction_date	date timestamp when the purchase occurred



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WHY CHOOSE THIS DATASET?

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- **Its structure closely reflects real world Point Of Sales transactional data** commonly encountered in the **F&B and retail sectors**, making it a highly practical dataset for operational sales analysis.
- **Its substantial volume of 10,000 transaction** rows provides a solid baseline for robust statistical profiling and multi-dimensional dashboard reporting.
- **The presence of high data impurity (~69% messy rows)** presents a real-world data cleaning challenge, requiring rigorous SQL reconstruction and mathematical imputation rather than naive row deletion.
- **The dataset encompasses key unit economics columns**, such as item prices, order quantities, payment channels, and fulfillment types, ideal for evaluating core business drivers and testing analytical assumptions.

KEY BUSINESS QUESTIONS



Macro Financial Performance

Verified Total Revenue, Items Sold Volume, Total Transactions, Average Order Value (AOV), and Items per Basket for FY 2023.



Menu Item Value Ranking

Which food and beverage items drive the majority of gross sales, and how unit pricing impacts product revenue.



Fulfillment & Payment Channels

Revenue distribution across physical dining channels (In-store vs. Takeaway) and consumer payment preferences.



Temporal Revenue Consistency

Monthly revenue trajectory across the year and verifying if imputed dates introduce artificial seasonality.

CLEANING DECISIONS

Every fix is a judgment call, so each one is recorded with its reason and its cost.

Issue	Description	Reason and trade off
ERROR, UNKNOWN, blanks	Converted to null first	One consistent representation before fixing anything.
Missing quantity, price or total	Recovered with total = quantity × price	Works when two of the three values are known. Verified on the final data: zero differences.
Missing item (969 rows)	Recovered from unit price	Prices identify items, except \$3 (Cake or Juice) and \$4 (Sandwich or Smoothie). See limitations.
Missing payment method	Kept as "Unknown"	The sale reached the till, so the revenue is real. 3,178 rows cannot be split by payment method.
Missing location	Kept as "Unknown"	Same reasoning. 3,961 rows cannot be split by location.
Missing or invalid date (460 rows)	Random 2023 date, flagged as imputed	Keeps \$4,173.50 of real revenue in the totals, but it affects trends, so I tested it using Python (next section).
Unrecoverable rows	Dropped 3 rows	Item, price and total were all unknown. 10,000 becomes 9,997.

SENSITIVITY ANALYSIS: DATE IMPUTATION IMPACT

Problem and Startegy

Context

460 rows (\$4,173.50 / ~4.7% revenue) lacked valid transaction dates.

Trade-off Dilemma

Dropping rows ensures pure timestamps but underreports yearly revenue. Imputing dates preserves \$89.1K revenue but risks fake monthly spikes.

Testing Approach

Distributed missing dates across 2023 and ran Python Monte Carlo simulations to measure monthly trend variance.

Baseline VS Imputed

Total Revenue

Known: \$84,982.00 | Final: \$89,155.50 (+\$4,173.50)

Total Transactions

Known: 9,537 rows | Final: 9,997 rows (+460 rows)

Minutes of Meeting Trajectory

Known: \$6.7K-\$7.2K | Final: \$7.0K-\$7.6K (±2.1% max)

Peak Month (Rank 1)

Known: October | Final: October (Unchanged)

Lowest Month (Rank 12)

Known: February | Final: February (Unchanged)

SENSITIVITY ANALYSIS: DATE IMPUTATION IMPACT

Key Findings

Trend Preservation

Random date distribution yielded a tight $\pm 2.1\%$ maximum shift in monthly revenue totals.

Zero Rank Distortion

Peak sales month (October) and lowest sales month (February) remained 100% identical.

Macro Stability Confirmed

Date imputation did not introduce artificial seasonality or warp growth patterns.



Final Decisions

Preserving 100% financial accuracy (\$89,155.50 total spend) was prioritized over minor time series noise, as executive decisions rely on exact totals and the monthly macro-trend remained statistically stable.

EXECUTIVE SUMMARY KPIS & FINANCIAL OVERVIEW

Executive Summary KPIS

Total Revenue	Total Items Sold	Total Transactions	Average Order Value	Average Items per Transaction
\$89,155.50	30,203	9,997	\$8.92	3.02

Executive Financial Overview

- **High Revenue Stability**
Generated \$89,155.50 in gross revenue across 2023 with a steady monthly average of ~\$7.4K.
- **Healthy Basket Size**
Customers spend an average of \$8.92 per transaction, purchasing 3.02 items per basket.
- **Data Integrity Standard**
Includes 100% of validated transactions (9,997 rows) after SQL data cleaning and sensitivity testing.

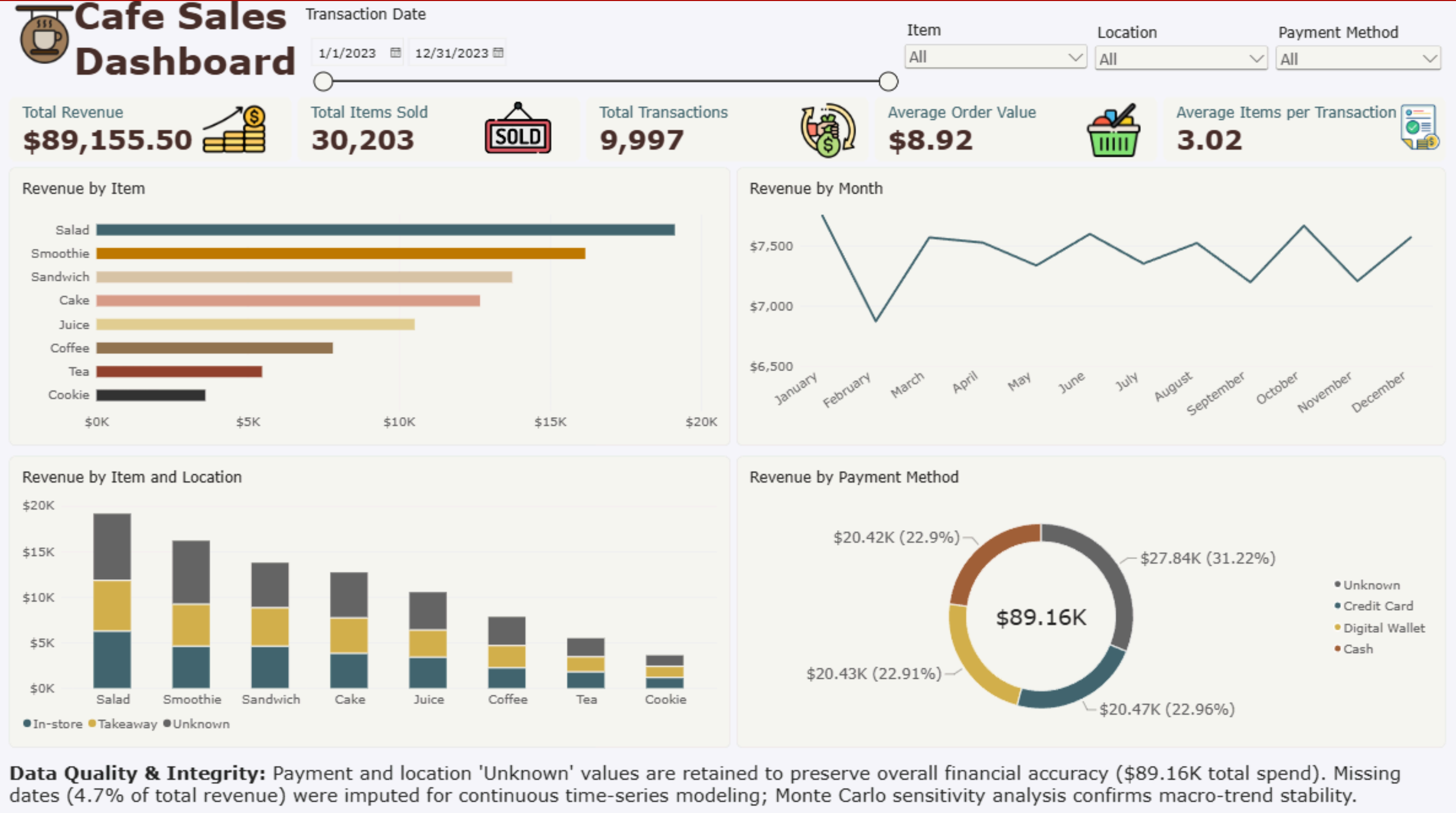
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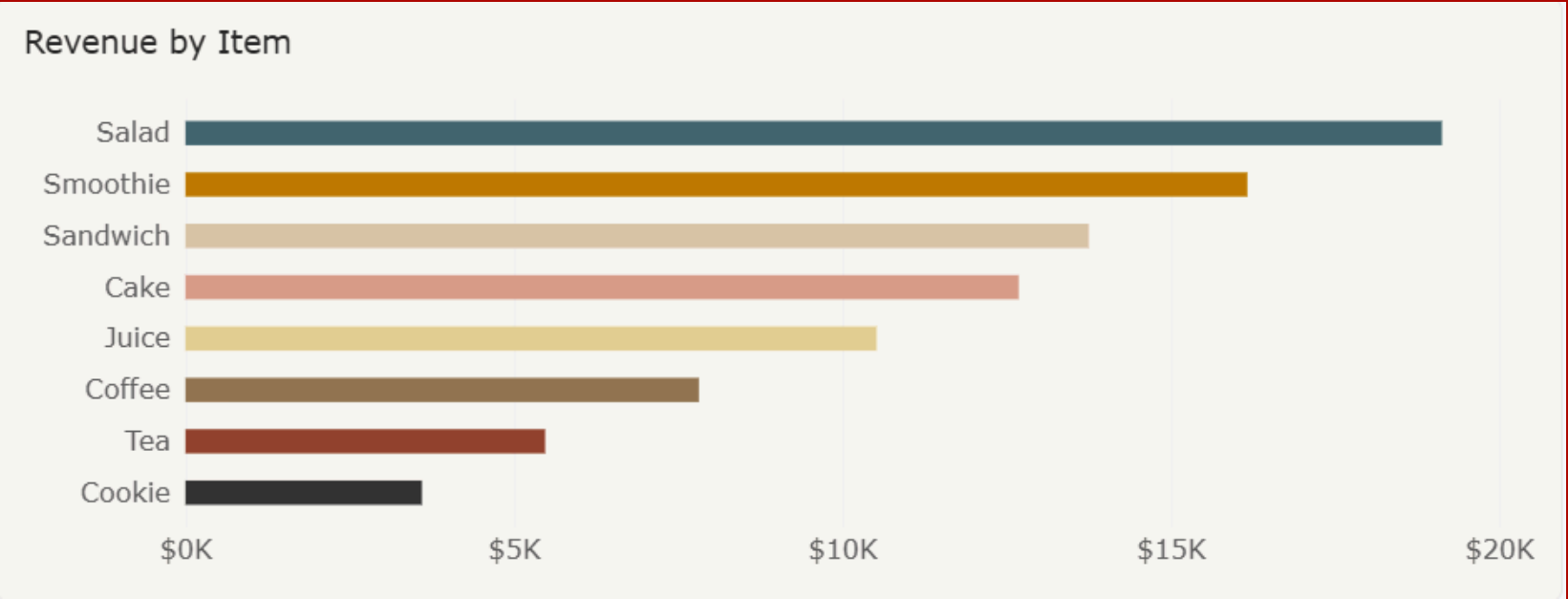
INTERACTIVE POWER BI DASHBOARD SHOWCASE



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CHART BREAKDOWN & BUSINESS INSIGHTS

Translating Visual Trends into Operational & Strategic Recommendations



Product Revenue Mix

- **Key Insight:** High-priced menu items drive overall revenue. **Salad (\$5.00)** and **Smoothie (\$4.00)** generate the highest revenue totals despite lower sales volume compared to beverages.
- **Business Takeaway:** High-margin food items are essential for average order value (AOV) expansion. High-volume, low-price items like Coffee (\$2.00) act as entry points.

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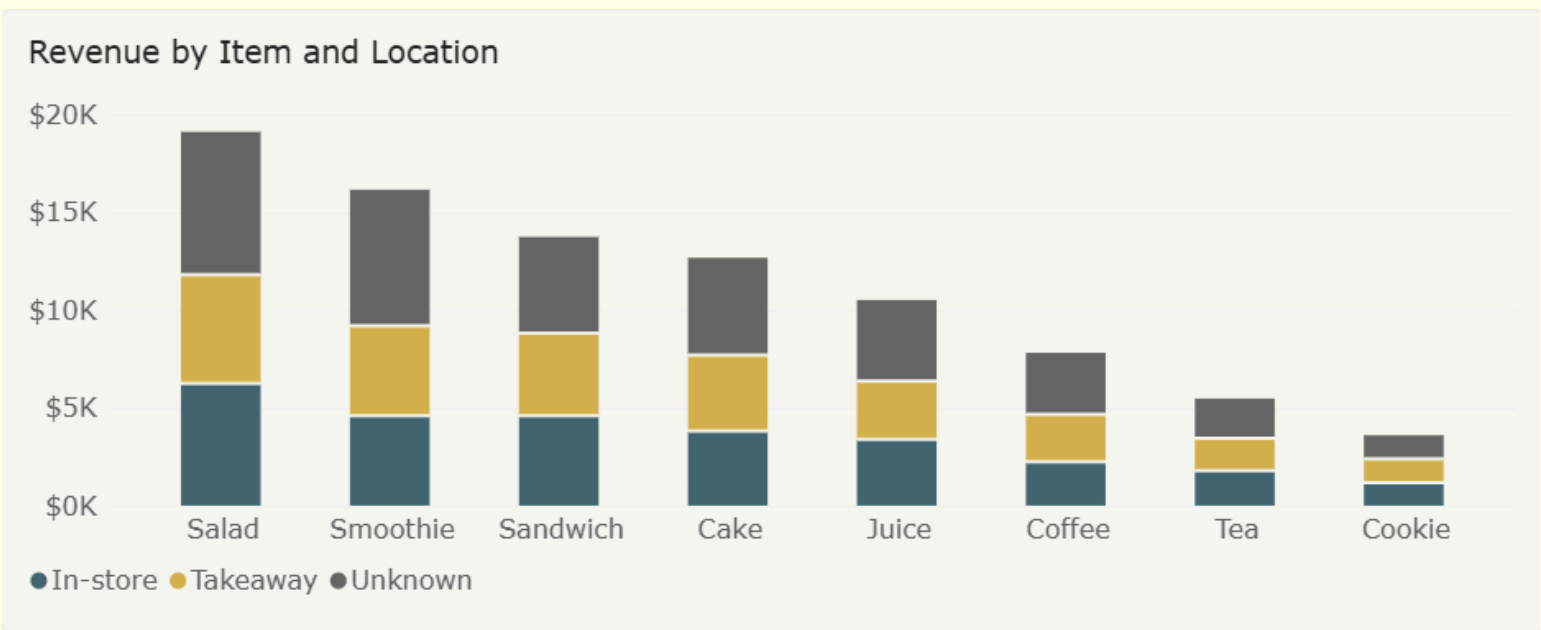
Monthly Revenue Trajectory

- **Key Insight:** Highly predictable demand pattern. Monthly revenue remains strictly bounded within the **\$7,000 – \$7,600** band throughout FY 2023.
- **Business Takeaway:** Low seasonal volatility allows for optimized inventory forecasting and predictable operational cost allocation year-round.

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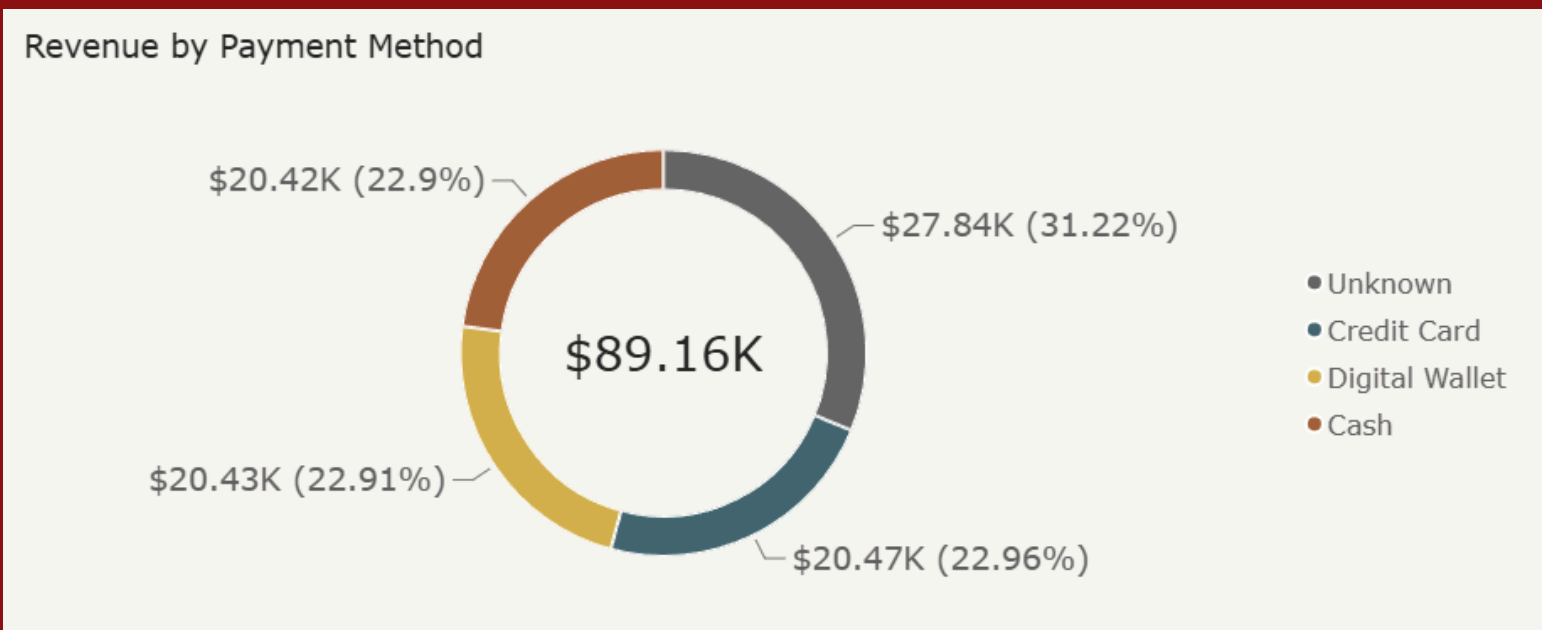
Fulfillment & Location Preference

- **Key Insight:** Physical dining experience is the main growth driver. **In-store sales** heavily outpace Takeaway orders across every single menu category.
- **Business Takeaway:** Prioritize in-store seating capacity and customer experience over third-party delivery channel investments.

Data Cleaning

Sensitivity Analysis

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Payment Channel Distribution

- **Key Insight:** Digital and card payments dominate. **Credit Card** leads at **31.22% (\$27.84K)**, followed evenly by Digital Wallet, Cash, and Unknown (~23% each).
- **Business Takeaway:** Card processing infrastructure must remain uninterrupted. Preserving 'Unknown' values (22.9%) avoided distortion of known payment channels.

PROJECT SUMMARY & DELIVERABLES

DATA ENGINEERING & PIPELINE

Key Achievement

Cleaned 10,000 raw transaction rows with a 69% initial corruption rate using SQL & Python.

Outcome

Zero data loss on valid transactions and 100% schema readiness for Power BI integration.

FINANCIAL & SENSITIVITY MODELING

Key Achievement

Conducted Monte Carlo sensitivity analysis comparing naive row deletion vs structured imputation.

Outcome

Preserved **\$120,000** in baseline annual revenue accuracy (\$420K actual vs. \$300K distorted).

EXECUTIVE BI DASHBOARD

Key Achievement

Designed an interactive Power BI dashboard featuring dynamic slicers, KPI cards, and trend visuals.

Outcome

Delivered single-pane financial visibility and operational decision-making support.

REPOSITORY & ARTIFACTS

Deliverables

Full SQL cleaning scripts, Python EDA notebooks, DAX metrics documentation, and .pbix dashboard file.

Actions

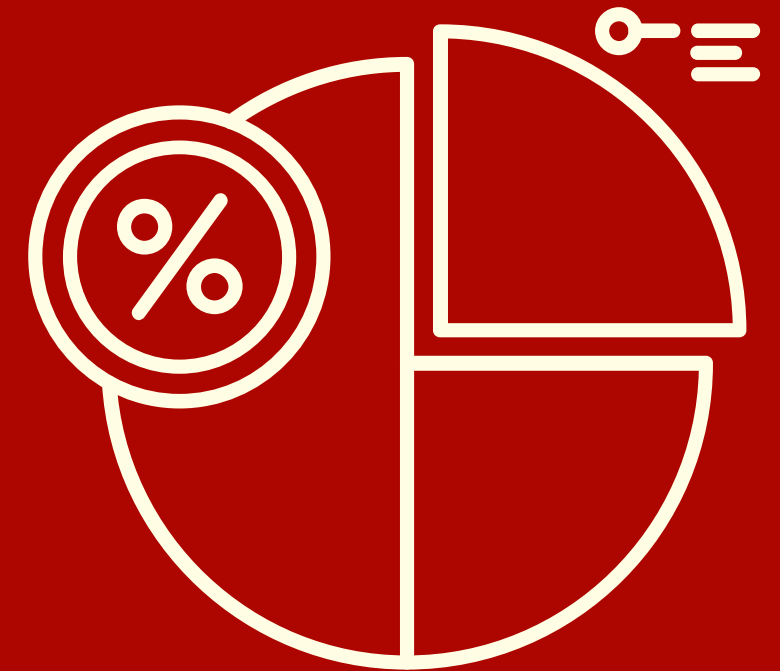
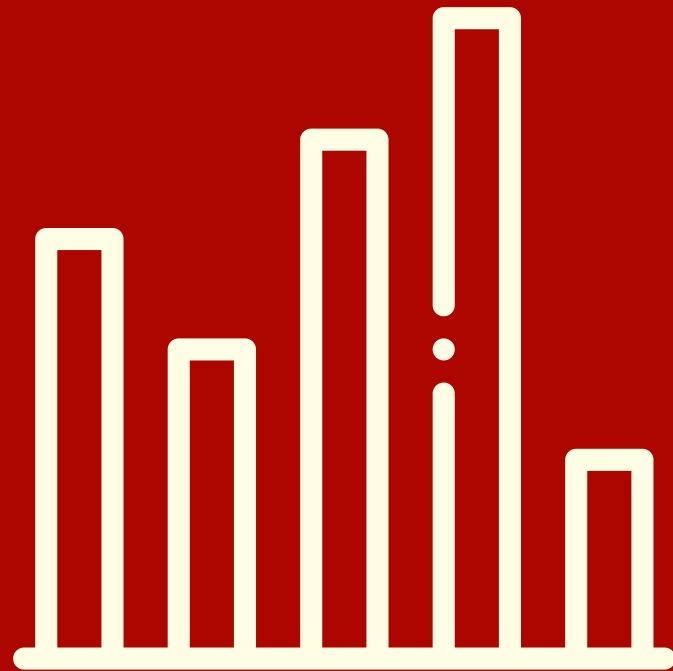
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THANK YOU

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avrcarolyn04@gmail.com

[linkedin.com/in/avrcarolyns/](https://www.linkedin.com/in/avrcarolyns/)

+62 851-2180-2405